

EE 615: Nano Electronics and Nano Microfabrication

Course Code	EE 615
Course Name	Nano Electronics and Nano Microfabrication
L-T-P-C	3-0-0-3
Prerequisites	IC 161-Applied Electronics, EE XX2 Microelectronics Circuits Design Practicum (MCDP), EE XX1 Electronics Devices, PH 501 Solid State Physics
Intended for	UG/MS/PhD
Distribution	Elective Semester : Odd/Even
Approval	6th Senate (May 2014)

Course Contents

- **Unit I:** Tunnel junction and applications of tunneling, Tunneling Through a Potential Barrier, Metal Insulator, Metal-Semiconductor, and Metal-Insulator-Metal Junctions, Coulomb Blockade, Tunnel Junctions, Tunnel Junction Excited by a Current Source. Spintronics and Foundations of nano-photonics. [6 Lectures]
- **Unit II:** Field Emission, GateOxide Tunneling and Hot Electron Effects in nano MOSFETs, Theory of Scanning Tunneling Microscope, Double Barrier Tunneling and the Resonant Tunneling Diode. [6 Lectures]
- **Unit III:** Introduction to lithography- Contact, proximity printing and Projection Printing, Resolution Enhancement techniques, overlay-accuracies, Mask-Error enhancement factor (MEEF), Positive and negative photoresists, Electron Lithography, Projection Printing, Direct writing, Electron resists. Lithography based on Surface Instabilities: Wetting, De-wetting, Adhesion, Limitations, Resolution and Achievable / line widths etc. Lift off process, Bulk Micro machining. [8 Lectures]
- **Unit IV:** Introduction to MEMS and NEMS, working principles, as micro sensors (acoustic wave sensor, biomedical and biosensor, chemical sensor, optical sensor, capacitive sensor, pressure sensor and thermal sensor), micro actuation (thermal actuation, piezoelectric actuation and electrostatic actuation) micro grippers, motors, valves, pumps, accelerometers, fluidics and capillary electrophoresis, active and passive micro fluidic devices, Piezoresistivity, Piezoelectricity and thermoelectricity, MEMS/NEMS design, processing, Oxidation, Sputter deposition, Evaporation, Chemical vapor deposition etc. [10 Lectures]
- **Unit V:** Introduction Scaling of physical systems Geometric scaling & Electrical system scaling. The Single-Electron Transistor: The Single- Electron Transistor Single-Electron Transistor Logic, Other SET and FET Structures, Carbon Nanotube Transistors (FETs and SETs), Semiconductor Nanowire FETs and SETs, Coulomb Blockade in a Nanocapacitor, Molecular SETs and Molecular Electronics. [10 Lectures]

Text Books

1. Stephen D. Senturia, **Microsystem Design**, Kluwer Academic Press
2. Marc Madou, **Fundamentals of microfabrication & Nanofabrication**.
3. T. Fukada & W. Mens, **Micro Mechanical system Principle & Technology**, Elsevier, 1998.
4. Julian W. Gardes, Vijay K. Varda, **Micro sensors MEMS & Smart Devices**, 2001.

References

1. WR Fahrner, **Nano Technology and Nano Electronics Materials, devices and measurement Techniques**, Springer.
2. T. Pradeep, **Nano: The Essentials Understanding Nano Science and Nanotechnology**, Tata McGraw Hill.
3. M. Ziese and M.J. Thornton, **Spin Electronics**
4. Karl Goser, Peter Glosekotter, Jan Dienstuhl, **Nanoelectronics and Nanosystems: From Transistor to Molecular and Quantum Devices**.
5. Shunri Odo and David Feny, **Silicon Nanoelectronics**, CRC Press.
6. C.N.R. Rao and A. Govindaraj, **Nanotubes and nanowires**, RSC Publishing.
7. M. Dutta and M.A. Stroscio, **Quantum-Based Electronic Devices and Systems**, World Scientific.
8. James R Sheats and Bruce w. Smith, **Microlithography Science and Technology**, Marcel Dekker, 1998.
9. J.P. Hirth and G.M. Pound, **Evaporation: Nucleation and Growth Kinetics**, Pergamon Press, 1963.